# SentinelNode

Low-cost ESP32 edge node with on-device TinyML for early detection of electronic component failures

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|---|---|
| Event | HackWithAditya 2026 — Level 1 online prototype screening |
| Track | Robotics, IoT & Smart Manufacturing |
| Team | Silicon Sentinels |
| Institution | Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad — B.Tech ECE, 3rd year |
| Members | Bankiti Srishanth Reddy (Team Lead) · Kothur Vedanidhi · L. Chandrapathi · Sanneella Ashmitha · Vasam Sai Varshini |
| Current status | Design complete; hardware build in progress. No bench or field results exist yet — every metric in this document is a target. |

**At a glance.** Electronic power and control boards in small factories fail without warning. SentinelNode is a ≈₹1,750 retrofit node that watches a board's rail current, ripple and component temperature, learns that board's healthy baseline on-device with an int8 TinyML autoencoder, and warns a technician (dashboard, SMS, LED/buzzer) before a part fails. **Mapping to screening criteria:** problem relevance §1–2 · solution and clarity §3–5 · prototype credibility §6 · validation §7 · feasibility §8 · originality §9.

## 1. Problem

SMPS power supplies, battery chargers, inverter/UPS boards and motor-drive/PLC control boards in small manufacturing units usually fail without warning. Maintenance is reactive: the fault is found only when the machine stops, and the board is then repaired or scrapped.

- **Heat is the dominant stressor.** A U.S. Air Force Avionics Integrity Program study, as reported by Siemens, attributes 55% of electronic system failures to temperature and 20% to vibration [R1].
- **Capacitors and power devices are the weak links.** About 30% of power-converter faults are attributed to capacitor degradation [R2]; an industry survey ranked power semiconductors the most fragile parts [R3].
- **Degradation is measurable before failure.** Electrolytic capacitor life roughly doubles per 10 °C reduction in temperature; as a capacitor ages its ESR rises, raising ripple and self-heating [R4, R5].
- **Downtime is expensive.** Siemens estimates unplanned downtime costs the 500 largest companies ≈11% of revenue (≈US$1.4 trillion/yr) [R6]. Small firms lose less in absolute terms, but each failure still halts production.

**The gap.** Tools for electronics health are benchtop thermal-imaging stations (e.g. FLIR ETS320 [R7]), heavy-industry vibration-led platforms (e.g. Infinite Uptime PlantOS [R8]) or research methods [R9, R10]. None is a low-cost device a micro or small enterprise can fit to an existing board to get an early warning without an expert. Telangana alone recorded ≈35.4 lakh Udyam/Udyam Assist registrations (Jul 2020 – Nov 2025) [R11].

## 2. Target users

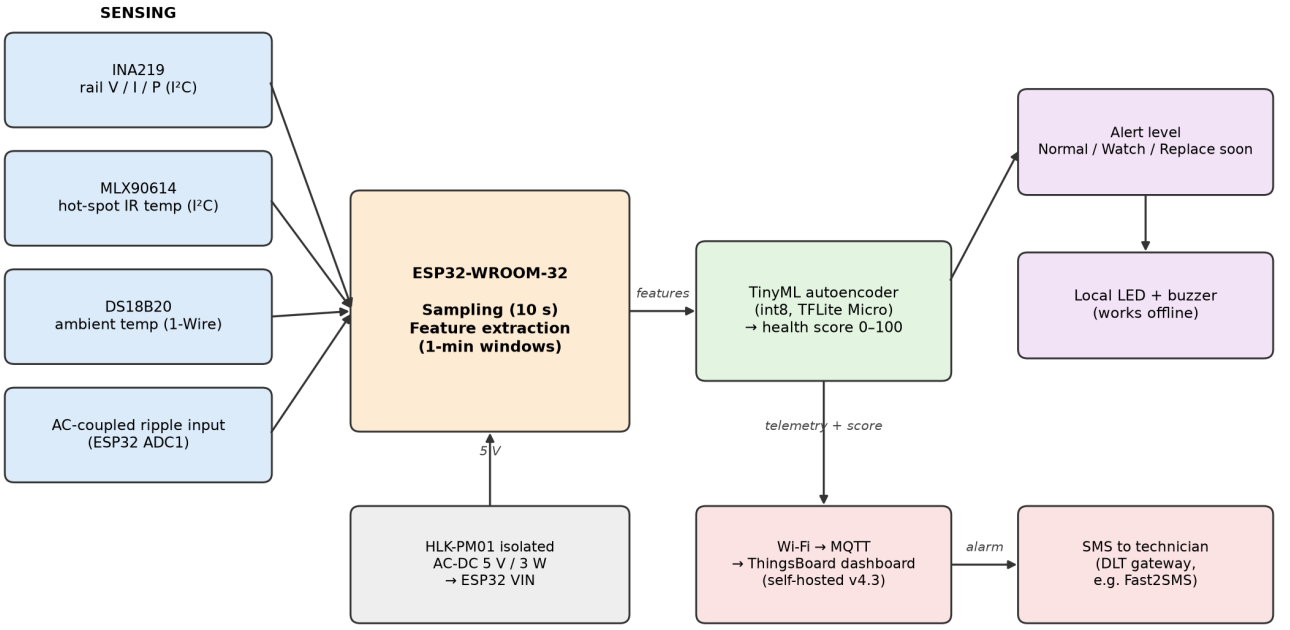
| User | Need | What SentinelNode gives them |
|---|---|---|
| Owners / maintenance technicians of micro and small manufacturing units | Avoid unplanned stoppages of power and control electronics | Early “Watch / Replace soon” alerts so a cheap part is swapped in a planned stop |
| Local electronics repair technicians | New recurring service income | Install nodes and offer a monitoring-plus-repair service |
| Institution and lab equipment managers | Keep lab power supplies and test rigs available | Continuous monitoring of shared equipment |
| ITI / polytechnic / engineering students | Hands-on IoT maintenance skills | Open hardware + firmware to build and study |

## 3. Solution — SentinelNode

A palm-sized node clips onto an existing low-voltage DC board (rails up to 26 V, e.g. 12/24 V SMPS outputs, chargers, control boards). It samples electrical and thermal indicators every 10 s, computes features over 1-minute windows, scores health 0–100 on the ESP32 itself, and raises a three-level alert: **Normal / Watch / Replace soon**. Alerts reach a dashboard and the technician’s phone by SMS; an on-board LED and buzzer keep working when Wi-Fi is down, so no alert depends on the cloud.

## 4. System architecture

**Figure 1 — SentinelNode system block diagram (Team Silicon Sentinels)**




*Design (prototype in development)*

Sensors on I²C (GPIO21/22), 1-Wire (GPIO4) and ADC1 (GPIO34); LED on GPIO25, buzzer on GPIO26.

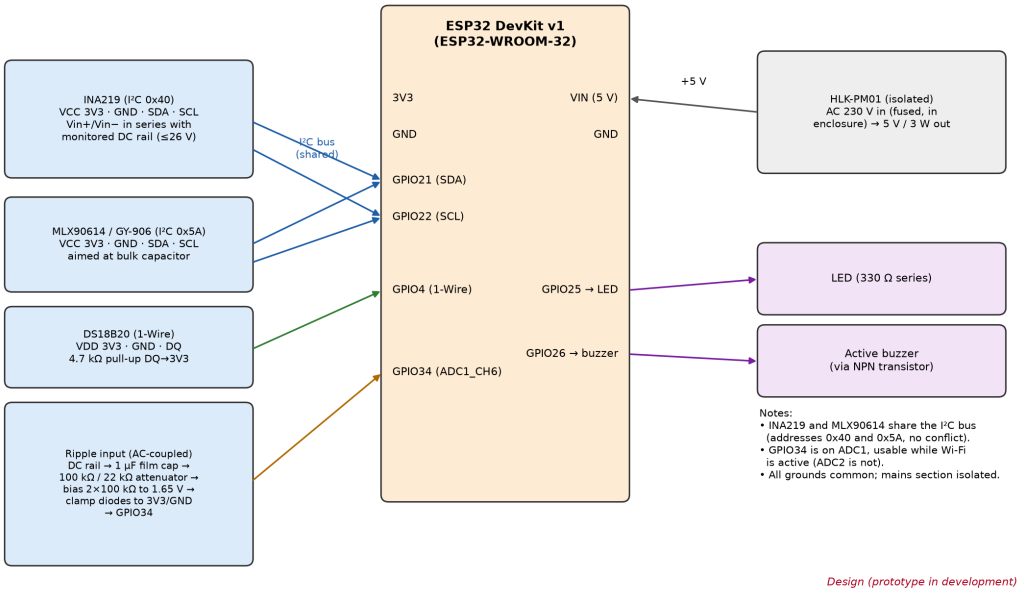
| Layer | Design |
|---|---|
| Sensing | INA219 (I²C 0x40): rail V, I, P · MLX90614 (I²C 0x5A): non-contact hot-spot temperature · DS18B20 (1-Wire): ambient reference · AC-coupled, attenuated ripple input to ESP32 ADC1 |
| Edge compute | ESP32-WROOM-32 (dual-core 240 MHz, Wi-Fi/BLE): sampling, feature extraction, int8 TinyML inference, alert logic |
| Communication | Wi-Fi → MQTT → self-hosted ThingsBoard dashboard [R21]; SMS via DLT-registered gateway, e.g. Fast2SMS [R22] |
| Local fallback | LED + buzzer alert levels, independent of network |
| Power / enclosure | HLK-PM01 isolated 5 V / 3 W AC-DC module; IP65 ABS enclosure 100×68×50 mm |

## 5. Hardware and bill of materials

All parts are off-the-shelf modules from Indian distributors. Prices are single-unit retail estimates including GST, checked October 2026; volume quotes have not been obtained.

| # | Item | Qty | Est. ₹ | Source |
|---|---|---|---|---|
| 1 | ESP32 DevKit v1 (ESP32-WROOM-32) | 1 | 322 | RobuKits [R16] |
| 2 | INA219 current/voltage/power module | 1 | 99 | Robu.in [R13] |
| 3 | MLX90614 (GY-906) non-contact IR sensor | 1 | 709 | Robu.in [R14] |
| 4 | DS18B20 waterproof temperature probe | 1 | 48 | Robu.in [R15] |
| 5 | HLK-PM01 5 V / 3 W isolated AC-DC module | 1 | 198 | Robu.in [R17] |
| 6 | IP65 ABS enclosure 100×68×50 mm | 1 | 121 | Price tracker [R18] |
| 7 | 2-layer PCB, passives, terminals, LED, buzzer, fuse | 1 | ≈250 | Team estimate; JLCPCB [R25] |
| | **Standard node total** | | **≈1,750** | |
| | Thermal variant: MLX90640 32×24 array (₹6,445 [R19]) replaces MLX90614 | | ≈7,490 | Whole-board hot-spot map |

**Figure 3 — Circuit-level connection diagram (pin mapping)**

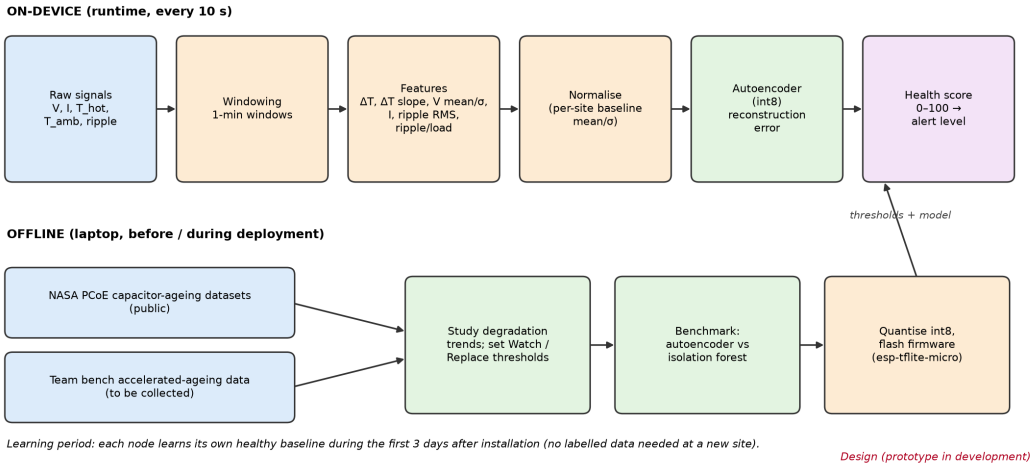



Planned wiring and pin map (design; not yet assembled).

Running cost ≈₹0–50 per node per month (self-hosted dashboard, site Wi-Fi, SMS at ₹0.25 each). Target installed price ₹2,500 — the node pays for itself if it prevents one failure costing more than that.

## 6. TinyML approach

**Figure 2 — Data flow and AI pipeline**




Team design; nothing in this pipeline has been run on hardware yet.

- **Features** (1-minute windows): temperature rise above ambient (ΔT) and its slope, rail-voltage mean and deviation, load current, RMS ripple, ripple-to-load ratio — load-normalised so normal load changes do not look like faults.
- **Model:** small dense autoencoder trained on each board's first days of healthy operation, quantised to int8 and run with TensorFlow Lite for Microcontrollers (Espressif esp-tflite-micro [R20]). Reconstruction error is mapped to a 0–100 health score with Watch / Replace-soon thresholds.
- **Baseline:** an isolation forest trained offline, to check the autoencoder earns its complexity.
- **Degradation knowledge:** NASA Ames PCoE capacitor accelerated-ageing datasets [R9, R10] are used offline to study degradation trends and set thresholds. Because they record impedance/capacitance, not our in-circuit signals, the team will also generate its own bench ageing data (§8).
- **Targets:** inference < 50 ms and model < 50 kB on the ESP32.

## 7. Prototype status and roadmap to the on-site build

**Honest status: design complete; hardware build in progress.** The architecture, part selection, pin map, feature set and model design above are final. No hardware has been tested and no measured results are claimed. At the on-site round we will demonstrate only what we have built and can explain line by line; our commit history will show the work progressing in the open.

| Milestone | Target date | Deliverable |
|---|---|---|
| M1 — Level 1 submission | by 20 Nov 2026 (deadline 25 Nov) | This document; design files and firmware skeleton in a public repository |
| M2 — Sensor bring-up | late Nov 2026 | INA219, MLX90614, DS18B20 and ripple channel streaming over serial and MQTT |
| M3 — Feature extraction + dashboard | early Dec 2026 | On-device features; ThingsBoard dashboard with live plots and alert levels |
| M4 — TinyML on device | mid Dec 2026 | int8 autoencoder running on the ESP32; isolation-forest baseline on a PC |
| M5 — On-site build (if shortlisted) | 18–20 Dec 2026, Aditya University | Integrate the node into the IP65 enclosure, add SMS alerts, run a live fault demo on a buck-converter board (e.g. heated / aged capacitor) and show the health score dropping |

**Ownership:** every member owns a block and can explain it at the viva — sensing and analog front-end, firmware and MQTT, TinyML training and quantisation, dashboard and alerts, enclosure/PCB and testing. Work is committed by its author so the history reflects who built what.

## 8. Validation plan (targets, not results)

| Step | Method | Target |
|---|---|---|
| 1. Sensor accuracy | Compare INA219 and MLX90614 with a bench multimeter and contact thermocouple | Current within ±2%; temperature within ±2 °C |
| 2. Offline benchmark | Train/test on NASA PCoE capacitor-ageing data; autoencoder vs isolation forest | Detect degradation onset before end-of-life criterion |
| 3. Accelerated ageing | Age 20 low-cost 105 °C electrolytic capacitors in a buck-converter rig at elevated ripple and temperature; LCR-meter ESR/capacitance as ground truth | Recall ≥ 85%; lead time ≥ 24 h; ≤ 1 false alarm per node per week on healthy controls |
| 4. On-device performance | Measure inference time and memory on the ESP32 | < 50 ms; < 50 kB |
| 5. Field pilot (after the hackathon) | 10 nodes on working SMPS/charger boards at 2–3 small units for ≥ 4 weeks; every alert verified against the maintenance log | Continuous operation; all alerts classified true/false |

## 9. Feasibility

- **Technical:** every block uses mature, documented modules and open toolchains (ESP-IDF/Arduino, esp-tflite-micro, ThingsBoard); no custom silicon.
- **Cost:** ≈₹1,750 per node; a 5-member student team can build several nodes within a small budget.
- **Time:** milestones M2–M4 fit the four weeks between the Level 1 deadline and the on-site round; the 48-hour build is reserved for integration and the live demo.
- **Skills:** the team’s ECE coursework covers embedded C, sensors, signals and basic ML.
- **Risks and mitigations:** false alarms after load changes → load-normalised features and per-board baselines; rails above 26 V → divider/shunt variant; mains safety → isolated supply and non-contact temperature sensing, installed by a trained person.

## 10. Originality versus existing solutions

| Existing approach | Limitation for our users | SentinelNode difference |
|---|---|---|
| IEEE 1856-2017 PHM framework [R23] | A standard, not a product | Implements sense → detect → diagnose → respond in a ≈₹1,750 retrofit node |
| NASA PCoE capacitor-ageing research [R9, R10] | Lab impedance instruments | Low-cost in-circuit proxies (ripple, ΔT, current) on a working board |
| Benchtop thermal imaging, e.g. FLIR ETS320 [R7] | Periodic, operator-driven inspection | Continuous, unattended, in-field monitoring |
| Industrial PdM, e.g. Infinite Uptime PlantOS [R8] | Heavy industry, rotating machinery, vibration-led | Built for electronic boards, priced for small units |
| Generic TinyML tools, e.g. Edge Impulse anomaly blocks [R24] | Toolkits, not a device | Ready open hardware + firmware with electronics-specific features and alert levels |

We apply established prognostics and health-management principles and do not claim to have invented PHM. What is new is the combination: an open, sub-₹2,000 retrofit node that learns each board’s own baseline on the device, uses electronics-specific degradation proxies (ΔT above ambient, ripple-to-load ratio), and alerts a local technician directly.

## 11. How this entry maps to the screening criteria

| Criterion | Where addressed | Key evidence |
|---|---|---|
| Problem relevance | §1, §2 | Cited failure and downtime data; clear small-industry users |
| Prototype credibility | §4–§7 | Final part list, pin map, model design; honest status; dated milestones; public commit history |
| Feasibility | §5, §8, §9 | Off-the-shelf ≈₹1,750 BOM; mature toolchains; risks with mitigations |
| Clarity | §3, Fig. 1–3 | One-page solution summary, diagrams, tables |
| Originality | §10 | Comparison with five existing approaches |
| Ownership (viva) | §7 | Each member owns a block and commits their own work |

## 12. References

R1 Siemens Simcenter blog, “Too hot – too shaky?” (citing USAF Avionics Integrity Program). blogs.sw.siemens.com/simcenter/too-hot-too-shaky-thermal-testing-and-vibration-control-testing-for-electronics/

R2 “An Overview of Condition Monitoring Techniques for Capacitors in DC-Link Applications,” IEEE Trans. Power Electronics, 2021. doi.org/10.1109/tpel.2020.3023469

R3 S. Yang et al., “An Industry-Based Survey of Reliability in Power Electronic Converters,” IEEE Trans. Ind. Appl., 47(3), 2011. doi.org/10.1109/tia.2011.2124436

R4 Cornell Dubilier, Aluminum Electrolytic Capacitor Application Guide. cde.com/resources/technical-papers/KNO_CD_AEappGuide_R2.pdf

R5 Cornell Dubilier, “Deriving Life Multipliers for Electrolytic Capacitors,” IEEE PELS Newsletter, 2004. cde.com/resources/technical-papers/multipliers.pdf

R6 Siemens, The True Cost of Downtime 2024. assets.new.siemens.com (tcod-2024.pdf)

R7 Teledyne FLIR, Electronic Benchtop Testing (ETS320). flir.com/instruments/science/ets/

R8 Infinite Uptime, PlantOS. infinite-uptime.com/prescriptive-ai-platform-plantos/

R9 NASA Ames PCoE Data Set Repository. nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/

R10 NASA Open Data, Accelerated Aging Experiments for Capacitor Health Monitoring and Prognostics. data.nasa.gov

R11 Ministry of MSME, Lok Sabha Unstarred Question No. 1883 (11.12.2025), Annexure-I (state-wise Udyam/UAP registrations).

R12 Invest Telangana, Electronics & Semiconductors (EMC Maheshwaram). invest.telangana.gov.in

R13–R19 Robu.in / RobuKits / price-tracker listings for INA219 (₹99), MLX90614 (₹709), DS18B20 (₹48), ESP32 DevKit v1 (₹322), HLK-PM01 (₹198), IP65 box (₹121), MLX90640 (₹6,445), checked Oct 2026.

R20 Espressif, esp-tflite-micro. github.com/espressif/esp-tflite-micro

R21 ThingsBoard open-source IoT platform. github.com/thingsboard/thingsboard

R22 Fast2SMS bulk SMS pricing. fast2sms.com/bulk-sms-pricing

R23 IEEE 1856-2017, Standard Framework for Prognostics and Health Management of Electronic Systems. standards.ieee.org/ieee/1856/5451/

R24 Edge Impulse, Anomaly detection (K-means). docs.edgeimpulse.com

R25 JLCPCB, PCB prototype pricing. jlcpcb.com